Question			Answer	Marks	AO	Guidance
7			**DR** Points of intersection with x-axis when $-3x^2 + 7x - 2 = 0$			
				M1	**3.1a**	Attempt to find intersection with x-axis
			$x = \frac{1}{3},\ 2$	**A1**	**1.1b**	Both exact roots seen
			Area $= \int_{\frac{1}{3}}^{2} \left(-3x^2 + 7x - 2\right) \mathrm{d}x$			
				M1*	**1.1a**	Allow for indefinite integral also
			$= \left[-x^3 + \frac{7}{2}x^2 - 2x\right]_{\frac{1}{3}}^{2}$	**A1**	**1.1b**	Correct indefinite integral
			$\left(-8 + \frac{7\times4}{2} - 2\times2\right) - \left(-\left(\frac{1}{3}\right)^3 + \frac{7}{2\times9} - \frac{2}{3}\right)$	**M1 (dep)**	**1.1b**	Substitution of their limits into their cubic expression must be seen
			$= \frac{125}{54}$	**A1**	**1.1b**	must be exact. Allow mixed number $2\frac{17}{54}$ or recurring decimal $2.31\dot{4}\dot{8}$ www
				[6]		